

DSAI501

Predicting Customer Churn for a Telecom Provider

Data Science Portfolio Assignment

Chosen problem statement Option A

Course How To Use Big-Data And AI To Make Better Business Decisions

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Executive Summary

This project tests whether customer account and service information can help identify telecom customers who are likely to leave. It uses the supplied Telco Customer Churn dataset, containing 7,043 customer records. The target is Churn, with 1,869 customers labelled Yes, giving a churn rate of 26.54%. Logistic regression and random forest are compared against a simple majority-class baseline. Data preparation, feature engineering and prediction are combined in a reusable pipeline.

Random forest was selected because its mean five-fold training F1 was 0.6316, just above logistic regression at 0.6280. On the untouched test set, the forest achieved 0.763 accuracy, 0.537 precision, 0.783 recall, 0.637 F1 and 0.843 ROC-AUC. Logistic regression achieved 0.734 accuracy, 0.499 precision, 0.789 recall, 0.611 F1 and 0.841 ROC-AUC. The dummy model reached 0.735 accuracy but zero recall and F1, showing why a majority prediction is not useful for retention. The main practical finding is that a reasonably accurate model still misses customers who churn and also flags customers who stay. A retention team could use its scores to prioritise a limited outreach trial, but these results do not establish that an offer will prevent churn. The dataset is an educational sample without the dated observations needed to prove advance prediction. A production decision would therefore require recent operational data, checks across customer groups and a controlled test of the proposed intervention.

Introduction

Customer churn means a customer stops using a company's service. For a subscription business, departures reduce future revenue and make customer relationships harder to maintain. A useful prediction could give a retention team time to investigate a problem or offer support. However, contacting every customer is expensive, and a discount given to someone who would have stayed anyway may reduce profit. The business question is therefore about directing limited attention, rather than simply obtaining the highest accuracy number.

The selected problem is Option A, Predicting Customer Churn for a Telecom Provider. The dataset is distributed by BlastChar on Kaggle and attributed to IBM sample data. Its description defines Churn as customers who left within the last month (BlastChar, n.d.). This project treats it as a public educational benchmark. It does not assume that these records represent current customers at a named operating company.

There are two related aims. The first is to compare two classification methods using the same preparation and evaluation procedure. The second is to explain how their predictions might support a sensible retention decision. Since the file has no observation dates, the immediate technical task is to classify the recorded churn label. Predicting departures before they happen remains the intended business use, but it needs stronger time-based evidence than this file can provide.

Data Description and Preprocessing

Each row represents one customer. The 21 original columns include an identifier, the churn target and 19 possible predictors. The predictors describe household characteristics, tenure in months, phone and internet services, contract type, payment method and charges. The supplied selected-columns file contains only the first ten columns and does not include Churn. It is therefore unsuitable for this supervised task on its own. The full CSV is used throughout, with the original file preserved for reproducibility.

The quality checks found no duplicated rows or customer identifiers. There were no missing values in the other fields, but TotalCharges contained 11 blank strings. All 11 belonged to customers with zero tenure. Converting this column to numeric turns these blanks into missing values. The records are retained because deleting new customers would remove a relevant customer segment. A median imputer fills the missing values using only the data available during each training fit. This is a simple, reproducible choice, although zero cumulative charges is another reasonable assumption that could be tested separately.

Tenure ranges from 0 to 72 months. MonthlyCharges ranges from 18.25 to 118.75, while recorded TotalCharges ranges from 18.80 to 8,684.80. These ranges did not justify arbitrary outlier removal. A high total charge can reflect a long relationship rather than an error. Categories such as No internet service and No phone service are kept as meaningful states, rather than treated as missing information. The customer identifier is excluded from prediction because it has no useful behavioural interpretation. Churn is encoded as 1 for Yes and 0 for No.

The split is made before fitting preprocessing. Numerical values are median-imputed and standardised, while categorical values are imputed with the most frequent training category and one-hot encoded. Encoding creates separate indicator columns instead of pretending that category names have a numerical order. Unseen categories can be handled at prediction time. The same preprocessing procedure sits inside every cross-validation fit, which prevents the validation rows from influencing fitted medians, scaling or category discovery (scikit-learn developers, n.d.-a).

Two engineered features are included. ServiceCount counts Yes across six optional internet services, from zero to six. It summarises how many services a customer uses, without replacing the original service fields. TenureBand groups customers into 0-12, 13-24, 25-48 and 49-plus months, allowing broad stages of the relationship to have different associations with churn. Both features use existing customer information and no target values. The training-fold ablation raised logistic regression F1 from 0.6275 with the base inputs to 0.6280 after adding the two engineered features. The change is small, so the features are retained as interpretable additions rather than presented as a major performance breakthrough. Their justification is therefore based on their meaning as well as the measured validation result, rather than an assumption that extra columns automatically improve a model.

Exploratory Data Analysis

Exploratory charts use the training portion, leaving test outcomes aside until the model choices are fixed. Figure 1 shows churn rates by contract. The full dataset contains more customers who stay than customers who leave, which immediately explains why accuracy alone is insufficient. A model that always predicts no churn can look fairly accurate while identifying nobody for retention. The dummy model makes that weakness visible in the final comparison.

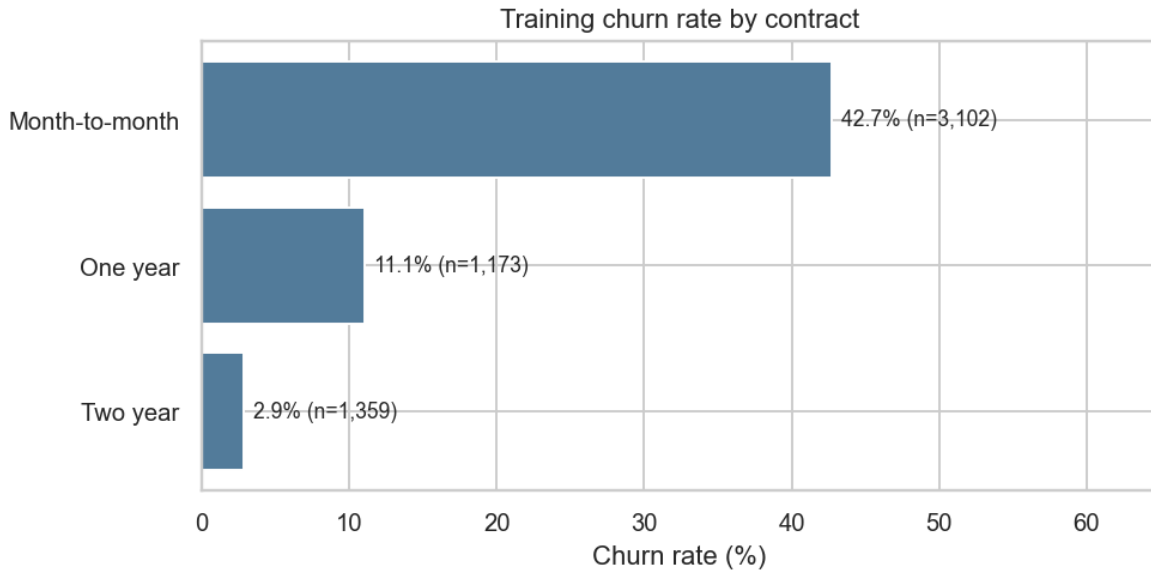


Figure 1. Training churn rate by contract type. Group sizes and rates are recorded in the notebook results.

In the training data, month-to-month customers had a 42.7% churn rate, compared with 11.1% for one-year contracts and 2.9% for two-year contracts. This suggests that contract information is useful for identifying risk. It does not establish that forcing a longer contract would reduce churn. Customers who expect to stay may choose longer commitments, while customers with uncertain needs may prefer flexibility. Pricing, satisfaction and circumstances could influence both decisions. The dataset does not separate these possible explanations.

Figure 2 compares tenure across the churn groups; the notebook also plots monthly charges. The median tenure of customers labelled No churn was 38 months, compared with 10 months for customers labelled Yes. Median monthly charges were 64.40 for the no-churn group and 79.95 for the churn group. The tenure pattern supports paying attention to the early customer experience, such as installation, onboarding and the first bills. The charge pattern could reflect affordability, service bundles or differences between internet plans. It should not be interpreted as evidence that cutting everyone's price would be profitable.

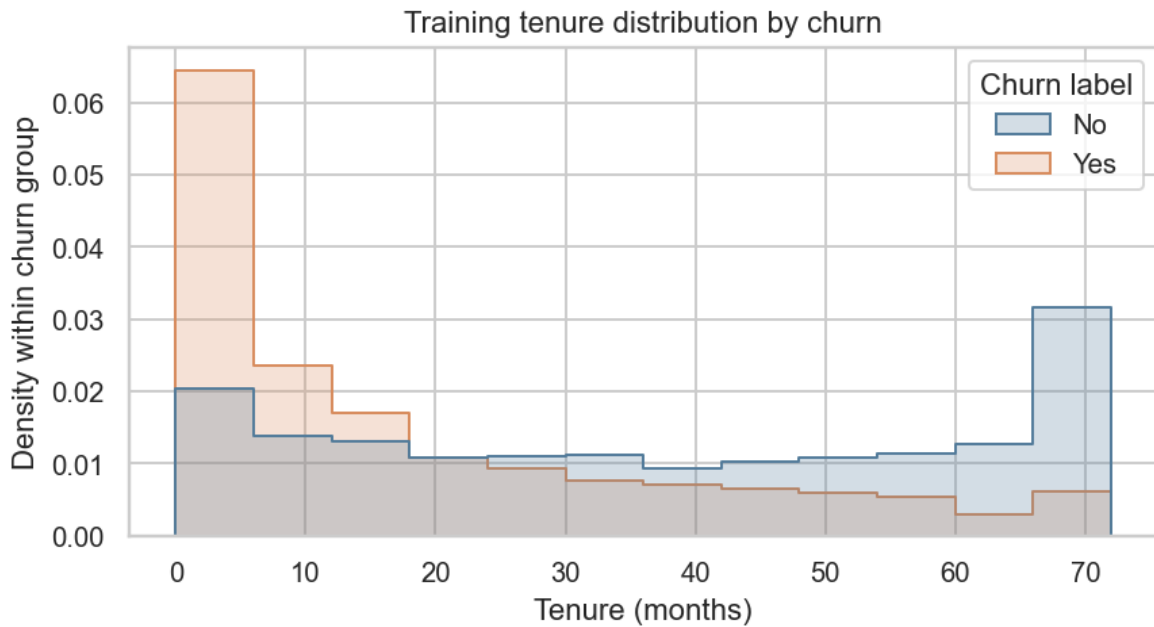


Figure 2. Training tenure distributions by observed churn label.

Figure 3 compares internet service categories. Fiber optic customers had a 42.1% churn rate, compared with 18.7% for DSL and 7.2% for customers without internet service. These figures are calculated on the training portion and are reported with group counts in the notebook output. The file does not contain outages, complaint histories or competitor offers, so it cannot explain the reason for this difference. These results are useful starting points for investigation. They describe associations in one sample, and the denominators matter when comparing customer groups. A large number of departures in a common category does not, by itself, imply a higher churn rate.

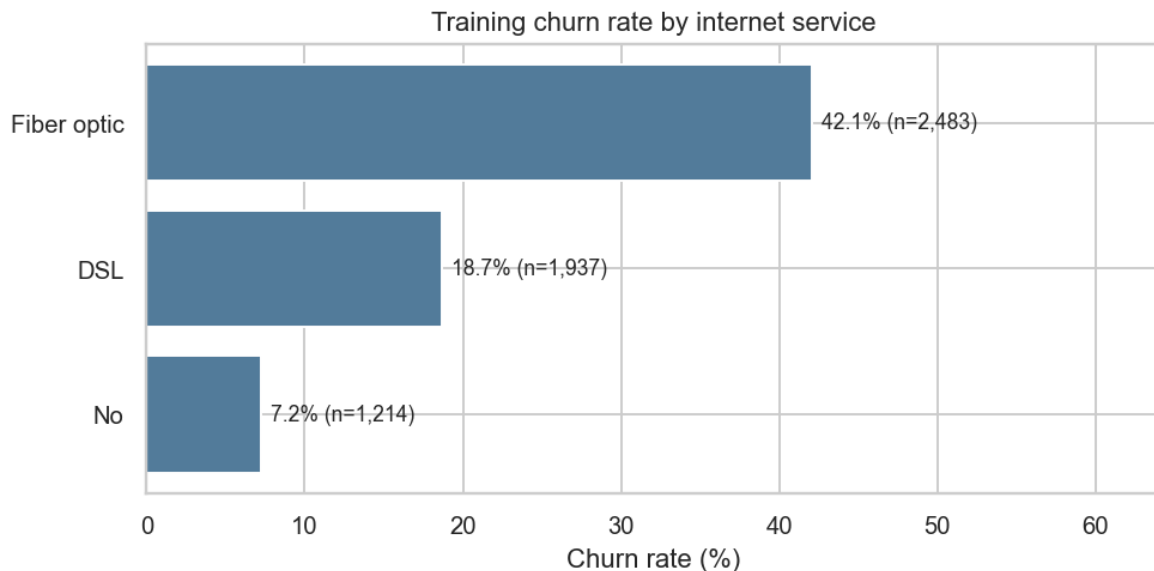


Figure 3. Training churn rate by internet service category.

Modelling Approach

The data is divided into 5,634 training customers and 1,409 test customers using an 80/20 stratified split and random state 42. Stratification keeps the churn proportions similar. Five-fold stratified cross-validation is then applied within the training set. Each candidate is fitted on four folds and evaluated on the remaining fold, with the process repeated across all five. The test set is reserved for the final comparison rather than used to choose settings (scikit-learn developers, n.d.-b).

Logistic regression estimates the probability of a binary outcome using a weighted combination of inputs. It provides a useful starting point because it is relatively simple and fast. Random forest combines many decision trees and can capture interactions or nonlinear patterns without specifying them in advance. Its flexibility may help, although it can also fit sample-specific detail. A DummyClassifier that always predicts the majority class provides the minimum comparison that the useful models should improve upon (scikit-learn developers, n.d.-c, n.d.-e).

The grid compares logistic regression values of $C = 0.1, 1$ and 10 with balanced or unweighted classes, and random forest variants with `max_depth` 8 or unlimited and balanced or unweighted classes. The best logistic regression used $C = 10$ and balanced weights. The best forest used `max_depth` 8, 200 trees, minimum leaf size 5 and balanced weights. Mean validation F1 is the model-selection criterion, chosen before inspecting test performance. F1 balances precision and recall when neither missed churners nor unnecessary contacts should be ignored. This is a practical compromise because the dataset contains no actual intervention costs. It is not a claim that F1 directly measures company profit. The same probability threshold of 0.5 is used for the reported classification comparisons, and the selection is not changed after seeing test results.

Accuracy measures the proportion of all predictions that are correct. Precision measures how many flagged customers actually churn, while recall measures how many actual churners are found. F1 is their harmonic mean and becomes low when either is poor. ROC-AUC evaluates how well the model ranks churners above non-churners across thresholds. It provides additional context, but does not measure whether individual probabilities are calibrated. All positive-class metrics refer to Churn equals Yes (scikit-learn developers, n.d.-d).

The exported model contains feature creation, preprocessing and the classifier together. It therefore accepts raw predictor columns without asking a user to recreate the engineered fields manually. The workflow reloads this file and compares its predictions with the in-memory model. Exact split membership, dependencies and output tables are also saved. These checks make the demonstration repeatable, although reproducibility within this environment does not establish that the model will generalise to a new company.

Results and Discussion

The held-out comparison is below. Random forest has the best F1 and accuracy among the trained models, while logistic regression has slightly higher recall. The difference is small enough that the simpler logistic model remains a reasonable alternative when explanation is prioritised. Table 1 reports all required test metrics, with ROC-AUC included as an extra measure. Cross-validation results are retained in the notebook so the final choice can be traced to training data. Small score differences should be interpreted cautiously because only one held-out split was used and no independent external dataset was available. A more complicated model is not automatically preferable when its practical improvement is small.

Model	Accuracy	Precision	Recall	F1	ROC AUC
Dummy	0.735	0.000	0.000	0.000	0.500
Logistic regression	0.734	0.499	0.789	0.611	0.841
Random forest	0.763	0.537	0.783	0.637	0.843

Table 1. Held-out test performance on 1,409 customers. Positive class is churn. The threshold is 0.5 for both trained classifiers.

For random forest, the confusion matrix is [TN, FP; FN, TP] = [782, 253; 81, 293]. It correctly leaves 782 non-churners unflagged and identifies 293 churners, while flagging 253 customers who stayed and missing 81 customers who churned. This is why the headline accuracy needs the confusion matrix in Figure 4 beside it. False negatives represent customers who could be missed by an outreach list. False positives represent customers who might receive an unnecessary call or offer. Their business costs depend on the action taken. A brief support message and an expensive discount should not be evaluated as if they cost the same amount.

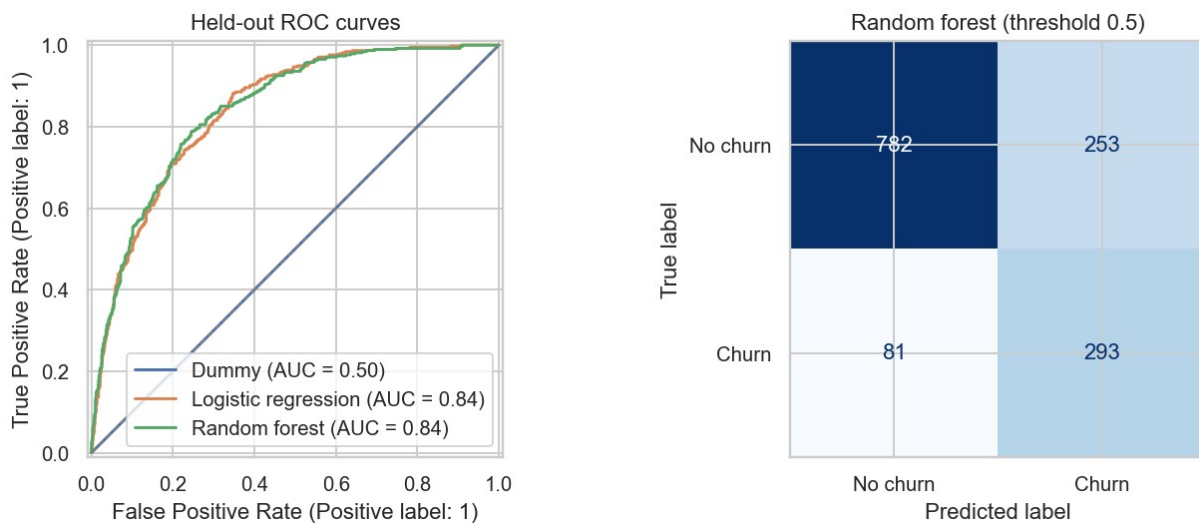


Figure 4. Held-out ROC curves and the selected random forest confusion matrix at threshold 0.50.

The post-hoc group table shows held-out recall of 75.6% for females and 81.2% for males, with false-positive rates of 23.9% and 25.0%. For SeniorCitizen = 0 and 1, recall is 74.6% and 88.8%, and false-positive rates are 21.8% and 43.5%. The senior-citizen group has 222 test records, so its rates are less stable. These checks are descriptive and do not certify fairness. These are descriptive checks after model selection, not proof that the model is fair or unfair. Different churn prevalence and small group counts affect the comparison. If risk scores determine access to discounts or support, the company should examine who receives help and who is missed. Removing demographic fields alone would not settle the issue, since other variables can carry related information. Group error rates should be monitored alongside the actual consequences of the retention policy.

Several limitations affect the interpretation. The sample lacks dated customer snapshots, so it cannot demonstrate that all features were available before a future departure. A random split estimates performance on similar records, not on later months. The file also lacks customer profit, complaints, competitor prices and intervention outcomes. Correlated features such as tenure and

total charges make a single-variable explanation incomplete. Model scores therefore indicate predictive associations rather than the effect of changing a contract or service.

Balanced class weights alter fitted scores, so the reported probabilities should be treated as risk scores until calibration is checked on representative validation data. The threshold of 0.5 is a consistent benchmark, not a verified business optimum. A lower threshold would generally produce more contacts and catch more churners, with additional false alarms. A real team could instead rank customers and contact only as many as its capacity permits. That rule should be selected using training or validation data and business costs, then tested on fresh observations. Repeatedly adjusting it against this test set would weaken the credibility of the reported evaluation.

Conclusion and Recommendations

Random forest is the selected model under the declared training-CV F1 rule. Logistic regression is close and may be preferred where a smaller, more transparent model is operationally useful. Contract type, tenure and service information provide useful signals for this benchmark. The results support a small, monitored retention trial rather than immediate deployment. Short-tenure customers could receive onboarding or service support, while flagged customers could be asked about specific difficulties before an incentive is offered. High risk alone does not tell the company which action will help.

For an illustrative campaign with capacity for 500 customers, the company could rank eligible customers by risk and review the top 500. The number is a planning example, not a measured result from this experiment. Customer value should mean expected retained contribution after service costs, rather than monthly revenue alone. A contact is worthwhile only when its expected incremental benefit exceeds its cost. The current dataset cannot supply the necessary treatment response or margin estimates.

The next step is to collect dated observations, define a future churn window and evaluate on a later period. A randomised pilot should compare contacted customers with an appropriate control group to estimate retention improvement and net cost. Calibration and errors across customer groups also need review. The project provides a working and explainable classification workflow, while these additional steps would determine whether it deserves a place in a real retention process.

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Assistance and attribution

The analysis uses Python (Python Software Foundation, n.d.), pandas (McKinney, 2010), NumPy (NumPy developers, n.d.), SciPy (SciPy developers, n.d.) and scikit-learn. Plots use Matplotlib (Hunter, 2007) and seaborn (Waskom, 2021). Joblib stores the model (Joblib developers, n.d.), and Jupyter provides the notebook format (Jupyter Project, n.d.). Document preparation uses python-docx and LibreOffice (python-docx contributors, n.d.; The Document Foundation, n.d.).

OpenAI Codex was used to assist with code revision, analysis execution, document drafting and checking. The supplied `churn_analysis.py` was the starting reference. Library methods and dataset sources are cited above. Group members should review the work and report their own contributions accurately under the course policy.